(i)(a) $x^2 + y^2 = 2ax$

$$2x + 2y\frac{dy}{dx} = 2a \implies a = x + y\frac{dy}{dx}$$

$$x^2 + y^2 = 2x\left(x + y\frac{dy}{dx}\right) = 2x^2 + 2xy\frac{dy}{dx}$$

$$\therefore \boxed{2xy\frac{dy}{dx} = y^2 - x^2}$$

$x^2 + y^2 = 2by$

$$2x + 2y\frac{dy}{dx} = 2b\frac{dy}{dx} \implies b = y + x\Big/\frac{dy}{dx}$$

$$x^2 + y^2 = 2y\left(y + x\Big/\frac{dy}{dx}\right) = 2y^2 + \frac{2xy}{\frac{dy}{dx}}$$

$$\therefore \boxed{\left(x^2 - y^2\right)\frac{dy}{dx} = 2xy}$$

(i)(b) At a common point with $y \neq x$, $x, y > 0$:

$$m_1 = \frac{y^2 - x^2}{2xy}, \quad m_2 = \frac{2xy}{x^2 - y^2}$$

$$m_1 m_2 = -1$$

$$\therefore \boxed{\text{the tangents are perpendicular}}$$

At (c, c): $a = c$ and $b = c$, giving circles centre $(c, 0)$ and $(0, c)$, both radius c. The radii to (c, c) are vertical and horizontal respectively

$$\therefore \boxed{\text{tangents: } y = c \text{ and } x = c \text{ — perpendicular}}$$

(ii) $y = c\ln x$

$$\frac{dy}{dx} = \frac{c}{x}, \quad c = \frac{y}{\ln x} \implies \frac{dy}{dx} = \frac{y}{x\ln x}$$

$$\frac{dy}{dx} = -\frac{x\ln x}{y}$$

$$\int y\,dy = -\int x\ln x\,dx$$

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x}{2} \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4}$$

$$\frac{y^2}{2} = -\frac{x^2}{2} \ln x + \frac{x^2}{4} + K$$

$$\therefore \boxed{y^2 = \tfrac{1}{2}x^2 - x^2 \ln x + C}$$

(iii) $y^2 = 4k(x+k)$

$$2y\frac{dy}{dx} = 4k \implies k = \tfrac{1}{2}y\frac{dy}{dx}$$

$$y^2 = 4 \cdot \tfrac{1}{2}y\frac{dy}{dx}\left(x + \tfrac{1}{2}y\frac{dy}{dx}\right) = 2xy\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2$$

Dividing by y^2 and writing $m = \dfrac{dy}{dx}$:

$$m^2 + \frac{2x}{y}m - 1 = 0$$

$$m_1 m_2 = -1$$

$$\therefore \boxed{\text{the tangents to the two curves are perpendicular}}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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